

Code for Additional Figures

This file contains the code for figures included in the paper but not in Models 1 or 2 (Figs 1, 2, and A.1-4). If downloading files, updated the “saveto” location before running.

Figure 1

Code for creating Fig 1. Note that this will not reproduce exactly the same figures between runs due to stochasticity. Subfigures were compiled in Adobe Illustrator 2024.

In[3698]:=

```
(*background functions for spectral synthesis and mimicry*)
cmplx[mod_, arg_] := ExpToTrig[mod Exp[I arg]];

SpecSynFourier[γ_, Nobs_, μ_ : 0, σ_ : 1, seed_ : 0] := Module[{
  phase, f, vec},
  If[seed == 0, SeedRandom[], SeedRandom[seed]];
  phase = RandomReal[{0, 2 Pi}, Nobs];
  f = Range[1 / Nobs, 1, 1 / Nobs];
  vec = InverseFourier[
    Join[{0 + 0 I}, Table[cmplx[1 / (f[[i]] ^ (γ / 2)), phase[[i]], {i, 1, (Nobs - 1)}]],
    FourierParameters → {-1, 1}];
  Standardize[Re[vec]] * σ + μ];

Mimic1[x_, z_] := Module[{y},
  y = x;
  Do [y[[Ordering[z, i] [[i]]] = x[[i]], {i, 1, Length[x]}];
  y];

(*TPC functions*)
gain[T_] := Exp[-((T - TI) ^ 2) / β];
loss[T_] := ma Exp[mb T] + mc;
r[T_] := gain[T] - loss[T];

(*example TPC parameters*)
TI = 29;
β = 180;
ma = 0.000005;
mb = 0.4;
mc = 0.05;
```

```
(*closed form solution of the r-alpha model*)
```

$$nt[t_, ntlast_] := \frac{r[Z[t]] \text{Exp}[r[Z[t]]]}{\left(\frac{r[Z[t]]}{ntlast} - \alpha\right) + \alpha \text{Exp}[r[Z[t]]]}$$

```
(*Subplot a: plot TPC against frequency distributions of temperatures*)
```

```
TPChist = Show[Plot[r[T] /.755, {T, 0, 37},
  PlotStyle -> {Thickness[0.02], Darker[Cyan]}, PlotRange -> {{0, 37}, {-0.15, 1.02}},
  Frame -> {True, True, False, True}, FrameStyle -> 24,
  FrameLabel -> {"Temperature, T (°C)", Style["TPC, r(T)", Darker[Cyan]],
    None, Style["Temperature Distributions", Gray]},
  LabelStyle -> Directive[Black], FrameTicks -> {Automatic, {0, 0.25, 0.5, 0.75, 1},
    None, Automatic}, ImageSize -> 500, AspectRatio -> 1,
  Epilog -> {Text[Style["(a)", Black, 30], {2, .97}],
    Text[Style["Average Fitness", Darker[Gray], 18], {10, .75}],
    Text[Style["Average Fitness", Purple, 18], {8, .64}],
    Text[Style["T1", Gray, 18], {23, .83}],
    Text[Style["T2, T3", Purple, 18], {26.5, .36}],

    Thickness[0.008], Gray, Line[{{23, -0.2}, {23, .797}}],
    Blue, Line[{{26, -0.2}, {26, .319}}],
    Red, Thickness[0.005], Line[{{26, -0.2}, {26, .319}}],

    Thickness[0.008], Dotted, Gray,
    Line[{{23 + 1, -0.2}, {23 + 1, .527}}], Line[{{23 - 1, -0.2}, {23 - 1, .527}}],
    Blue, Line[{{26 + 4, -0.2}, {26 + 4, .088}}],
    Line[{{26 - 4, -0.2}, {26 - 4, .088}}],
    Red, Thickness[0.005],
    Line[{{26 + 4, -0.2}, {26 + 4, .088}}], Line[{{26 - 4, -0.2}, {26 - 4, .088}}],

    Dashed, Darker[Gray],
    {Dashed, Darker[Red]}, Line[{{0, .7125}, {28.3, .7125}}],
    Darker[Blue], Line[{{0, .6008}, {28.4, .6008}}],
    Darker[Red], Line[{{0.3, .6008}, {28.4, .6008}}]},
  Plot[2 * PDF[NormalDistribution[23, 1], x], {x, 0, 37},
  PlotStyle -> {Thickness[0.01], Gray}, PlotRange -> Full, Filling -> Axis,
  Frame -> {True, False, False, True}, FrameLabel -> {None, None, None, "Frequency"},
  LabelStyle -> Directive[Black], ImageSize -> 500],
  Plot[2 * PDF[NormalDistribution[26, 2.5], x],
  {x, 0, 37}, PlotStyle -> {Thickness[0.01], Blue},
  Filling -> Axis, PlotRange -> {{0, 37}, Automatic (*{0, 320}*)},
  Frame -> {True, False, False, True}, FrameLabel -> {None, None, None, "Frequency"},
  LabelStyle -> Directive[Black], ImageSize -> 500],
```

```

Plot[2 * PDF[NormalDistribution[26, 2.5], x],
  {x, 0, 37}, PlotStyle → {Thickness[0.01], DotDashed, Red},
  Filling → Axis, PlotRange → {{0, 37}, Automatic (*{0, 320}*)},
  Frame → {True, False, False, True}, FrameLabel → {None, None, None, "Frequency"},
  LabelStyle → Directive[Black], ImageSize → 500]]

(*subplots b, c, d,
e: plot time series of temperatures and abundances for 3 time series*)
 $\alpha$  = .001; (*density dependence parameter*)
tmax = 1000; (*number of time steps*)
maxsims = 1; (*number of replicates*)
 $\gamma$ vals = Range[0, 2, 2]; (*Autocorrelation level: 0, 2*)
 $\mu$ Tvals = Range[23, 26, 3]; (*Mean temp: 23, 26*)
 $\sigma$ Tvals = Range[1, 2.5, 1.5]; (*St dev of temp: 1, 2.5*)

(*create temperature time series*)
W = ConstantArray[0, {Length[ $\mu$ Tvals], Length[ $\sigma$ Tvals], tmax}];
Do[W[[i, j, k]] = InverseCDF[NormalDistribution[ $\mu$ Tvals[[i]],  $\sigma$ Tvals[[j]]], k * (1 / tmax)],
  {i, 1, Length[ $\mu$ Tvals]}, {j, 1, Length[ $\sigma$ Tvals]}, {k, 1, tmax - 1}]
Do[W[[i, j, tmax]] =
  InverseCDF[NormalDistribution[ $\mu$ Tvals[[i]],  $\sigma$ Tvals[[j]]], (tmax - .5) * (1 / tmax)],
  {i, 1, Length[ $\mu$ Tvals]}, {j, 1, Length[ $\sigma$ Tvals]}];

(*calculate initial population sizes (=the avg predicted by the distribution)*)
ninit = ConstantArray[0, {Length[ $\mu$ Tvals], Length[ $\sigma$ Tvals]}];
Do[ninit[[i, k]] = (Mean[Table[r[W[[i, k, j]]], {j, 1, tmax}]]]) /  $\alpha$ ,
  {k, 1, Length[ $\sigma$ Tvals]}, {i, 1, Length[ $\mu$ Tvals]}]
lt23 = ConstantArray[ninit[[1, 1]], tmax];
lt26 = ConstantArray[ninit[[2, 2]], tmax];

(*initialize array to store extinction times; =tmax if not extinct*)
extTime =
  ConstantArray[tmax, {Length[ $\gamma$ vals], Length[ $\mu$ Tvals], Length[ $\sigma$ Tvals], maxsims}];
(*initialize array to store population sizes*)
p1 = ConstantArray[0,
  {Length[ $\gamma$ vals], Length[ $\mu$ Tvals], Length[ $\sigma$ Tvals], maxsims, tmax}];
Do[p1[[i, j, m, l, 1]] = ninit[[j, m]], {l, 1, maxsims},
  {m, 1, Length[ $\sigma$ Tvals]}, {j, 1, Length[ $\mu$ Tvals]}, {i, 1, Length[ $\gamma$ vals]}]

(*create array to store temperature time series*)
temps1 = ConstantArray[0, {Length[ $\mu$ Tvals], Length[ $\sigma$ Tvals], Length[ $\gamma$ vals]}];

(*run simulation*)

```

```

SetSharedVariable[p1, extTime];
Do[ $\gamma$  =  $\gamma$ vals[[i]];
 $\mu$ T =  $\mu$ Tvals[[j]];
 $\sigma$ T =  $\sigma$ Tvals[[m]];

X = SpecSynFourier[ $\gamma$ , tmax];
Z = Mimic1[W[[j, m]], X];
envt = Interpolation[Z, InterpolationOrder  $\rightarrow$  0];

temps1[[i, j, m]] = Z;

Do[p1[[i, j, m, l, k]] = nt[k, p1[[i, j, m, l, k - 1]]];
If[p1[[i, j, m, l, k]] < 1, extTime[[i, j, m, l]] = k;
Break[]];, {k, 2, tmax}],

{l, 1, maxsims}, {m, 1, Length[ $\sigma$ Tvals]},
{j, 1, Length[ $\mu$ Tvals]}, {i, 1, Length[ $\gamma$ vals]]}

(*generate subfigures*)
framesize = 20;
sublabel = 24;

series1 = ListLinePlot[{temps1[[1, 1, 1]]}, PlotStyle  $\rightarrow$  {Gray, Blue, Red},
PlotRange  $\rightarrow$  {{0, tmax}, {17.5, 35}}, Frame  $\rightarrow$  {True, True, False, False},
FrameStyle  $\rightarrow$  framesize, FrameLabel  $\rightarrow$  {"Time", "Temperature"},
LabelStyle  $\rightarrow$  Directive[Black], ImageSize  $\rightarrow$  500, AspectRatio  $\rightarrow$  .3,
Epilog  $\rightarrow$  {{Text[Style["(b)", Black, sublabel], {45, 33.5}]},
Thickness[0.003], Black, Line[{{0, 23}, {tmax, 23}}],
Dashed, Line[{{0, 23 + 1}, {tmax, 23 + 1}}], Line[{{0, 23 - 1}, {tmax, 23 - 1}}],
Dotted, Black,
Line[{{0, 26.29}, {tmax, 26.29}}], Line[{{0, 19.91}, {tmax, 19.91}}],
Dotted, Blue,
Line[{{90, 34.2}, {tmax, 34.2}}], Line[{{0, 18.3}, {tmax, 18.3}}],
Dotted, Darker[Red],
Line[{{95, 34.2}, {tmax, 34.2}}], Line[{{5, 18.3}, {tmax, 18.3}}]}]
series2 = ListLinePlot[{temps1[[1, 2, 2]]}, PlotStyle  $\rightarrow$  {Blue},
PlotRange  $\rightarrow$  {{0, tmax}, {17.5, 35}}, Frame  $\rightarrow$  {True, True, False, False},
FrameStyle  $\rightarrow$  framesize, FrameLabel  $\rightarrow$  {"Time", "Temperature"},
LabelStyle  $\rightarrow$  Directive[Black], ImageSize  $\rightarrow$  500, AspectRatio  $\rightarrow$  .3,
Epilog  $\rightarrow$  {{Text[Style["(c)", Black, sublabel], {45, 33.5}]},
Thickness[0.003], Black, Line[{{0, 26}, {tmax, 26}}],
Dashed, Line[{{0, 26 + 3}, {tmax, 26 + 3}}], Line[{{0, 26 - 4}, {tmax, 26 - 3}}],
Dotted, Blue,
Line[{{90, 34.2}, {tmax, 34.2}}], Line[{{0, 18.3}, {tmax, 18.3}}],

```

```

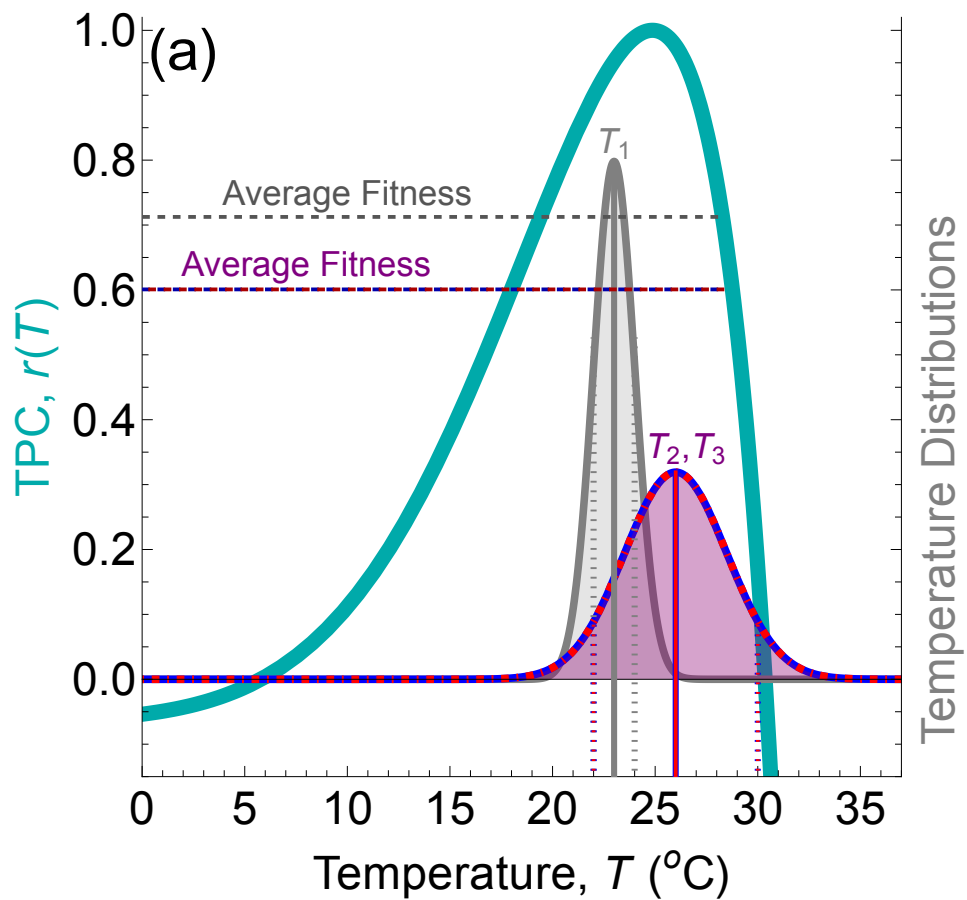
Dotted, Darker[Red],
Line[{{95, 34.2}, {tmax, 34.2}}], Line[{{5, 18.3}, {tmax, 18.3}}]]]
series3 = ListLinePlot[{temps1[[2, 2, 2]]}, PlotStyle → {Red},
PlotRange → {{0, tmax}, {17.5, 35}}, Frame → {True, True, False, False},
FrameStyle → framesize, FrameLabel → {"Time", "Temperature"},
LabelStyle → Directive[Black], ImageSize → 500, AspectRatio → .3,
Epilog → {{Text[Style["(d)", Black, sublabel], {45, 33.5}]},
Thickness[0.003], Darker[Red], Line[{{0, 26}, {tmax, 26}}],
Dashed, Line[{{0, 26 + 3}, {tmax, 26 + 3}}], Line[{{0, 26 - 3}, {tmax, 26 - 3}}],
Dotted, Blue,
Line[{{90, 34.2}, {tmax, 34.2}}], Line[{{0, 18.3}, {tmax, 18.3}}],
Dotted, Darker[Red],
Line[{{95, 34.2}, {tmax, 34.2}}], Line[{{5, 18.3}, {tmax, 18.3}}]]]

data = {p1[[1, 1, 1, 1]], p1[[1, 2, 2, 1]], p1[[2, 2, 2, 1]], lt23, lt26, lt26};
abundances = ListLinePlot[data, ScalingFunctions → {"Linear", "Log"},
PlotStyle → {Gray, {Thickness[0.004], Blue}, {Thickness[0.01], Red},
{Dashed, Black}, Darker[Blue], {Dashed, Darker[Red]}},
PlotLegends → Placed[{" $\mu = 23$ ,  $\sigma = 1$ ,  $\gamma = 0$ ", " $\mu = 26$ ,  $\sigma = 3$ ,  $\gamma = 0$ ",
" $\mu = 26$ ,  $\sigma = 3$ ,  $\gamma = 2$ "}, {0.8, 0.2}],
PlotRange → {{0, tmax}, {1, 1800}}, Frame → {True, True, False, False},
FrameStyle → 16, FrameLabel → {"Time", "Abundance"},
LabelStyle → Directive[Black], ImageSize → 500, AspectRatio → .8,
Epilog → {Text[Style["(e)", Black, 28], Scaled[{0.05, 0.97}]]}]

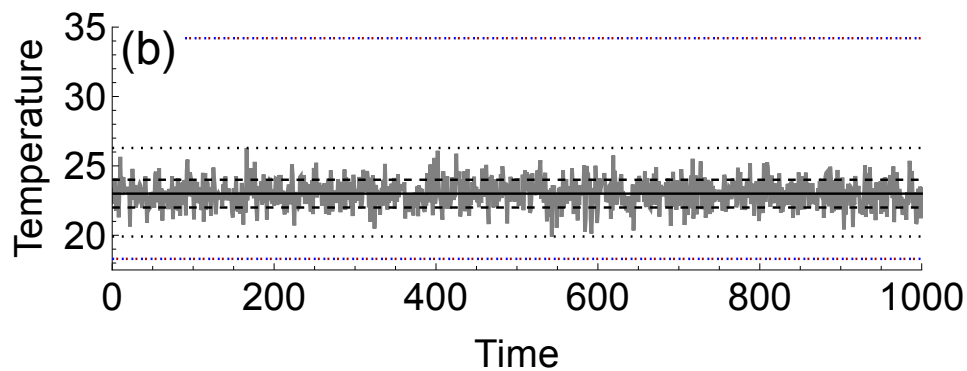
(*saveto="/EXPORTLOCATION/";
Export[saveto<>"tpcHist.pdf",TPChist,ImageResolution→1000];
Export[saveto<>"series1.pdf",series1,ImageResolution→1000];
Export[saveto<>"series2.pdf",series2,ImageResolution→1000];
Export[saveto<>"series3.pdf",series3,ImageResolution→1000];
Export[saveto<>"abundances.pdf",abundances,ImageResolution→1000];*)

```

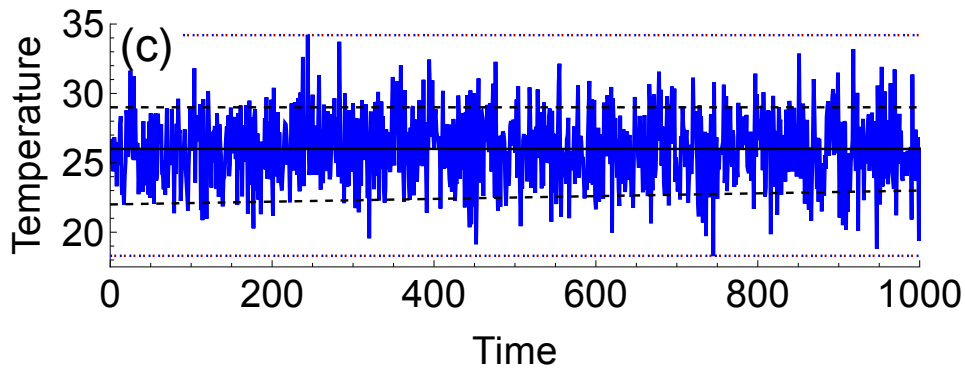
Out[3710]=



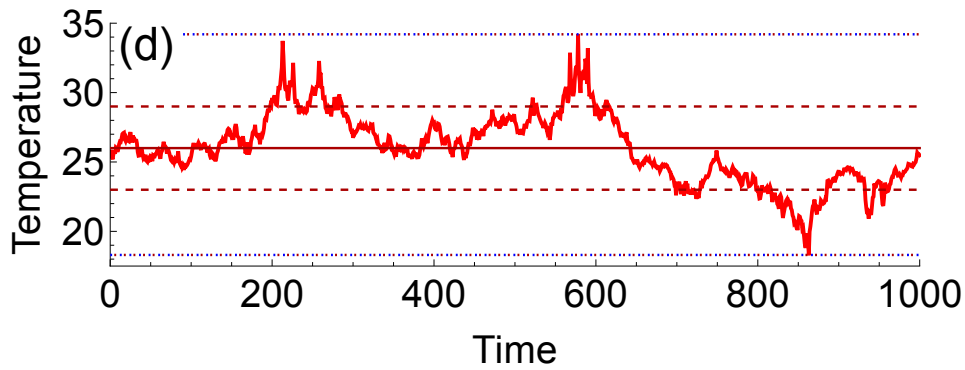
Out[3732]=



Out[3733]=



Out[3734]=



Out[3736]=

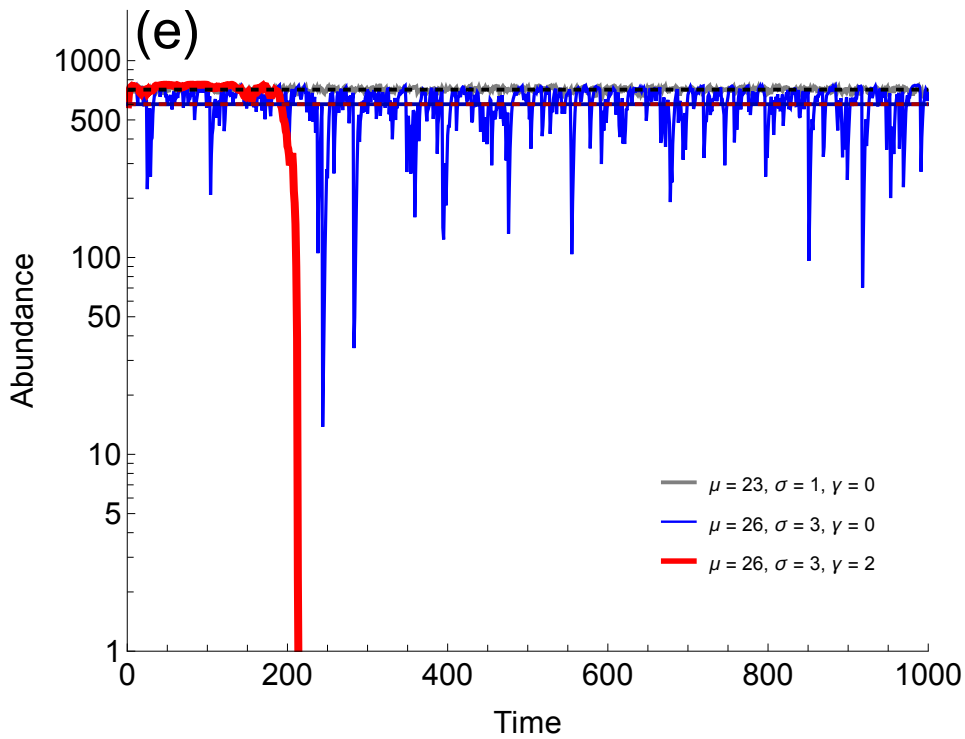


Figure 2

This code creates a figure of the generic TPCs used in the initial models and the corresponding thermal distributions (Fig 2).

In[3562]:=

```
(* Parameters of the TPC model (generalist, then specialist)*)
TI = 29; (*move bopt*)
 $\beta$  = 180; (*make narrower*)
ma = 0.000005;
mb = 0.4;
mc = 0.05;
TI1 = 35;
 $\beta$ 1 = 170;

gencol = Darker[Cyan];
specol = Darker[Purple];

gopt = 25;
sopt = 28;
gmin = 5.8;
smin = 14.54;
gmax = 30.4;
smax = 32;

(* Plot*)
TPCs = Show[Plot[r[T] /. 755, {T, 0, 37}, Filling -> Axis,
  PlotStyle -> gencol, PlotLegends -> Placed[{"generalist"}, {0.185, 0.8}],
  PlotRange -> {{2, 37}, {-0.2, 1.1}}, Frame -> {True, True, False, False},
  FrameStyle -> 14, FrameLabel -> {"Temperature, T (°C)",
    "Normalized Malthusian Fitness, r (time-1)"}, LabelStyle -> Directive[Black],
  FrameTicks -> {Automatic, {0, 0.25, 0.5, 0.75, 1}}, ImageSize -> 500],
Plot[(r[T - 2, TI1,  $\beta$ 1, ma, mb, mc]) /. 4067, {T, 0, 37}, Filling -> Axis,
  PlotStyle -> specol, PlotLegends -> Placed[{"specialist"}, {0.182, 0.7}]],
Epilog -> {Text[Style["(a)", Black, 18], {3.3, 1.05}],
  {Text[Style["Topt = 25", gencol], {22.8, 1.02}]},
  {Text[Style["Tmin = 5.8", gencol], {4, .04}]},
  {Text[Style["Tmax = 30.4", gencol], {28, -.05}]},
  {Text[Style["Topt = 28", specol], {30.2, 1.02}]},
  {Text[Style["Tmin = 14.5", specol], {17, -.05}]},
  {Text[Style["Tmax = 32", specol], {34, .04}]},
  {Black, PointSize[0.015], Point[{gmin, 0}]},
  {Black, PointSize[0.015], Point[{smin, 0}]},
  {Black, PointSize[0.015], Point[{gopt, 1}]},
  {Black, PointSize[0.015], Point[{sopt, 1}]},
```

```

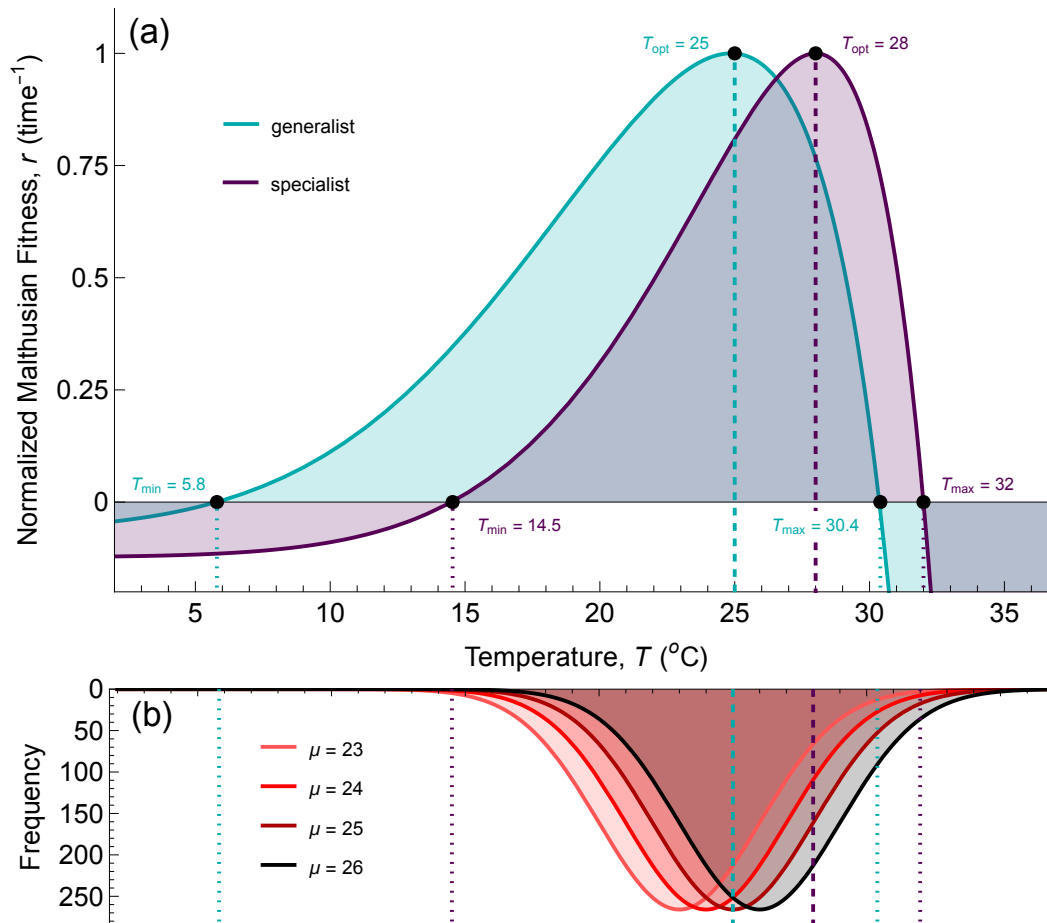
{Black, PointSize[0.015], Point[{gmax, 0}]},
{Black, PointSize[0.015], Point[{smax, 0}]},
Thick, specol, Dotted, Line[{smax, -0.015}, {smax, -.25}],
Line[{smin, -0.015}, {smin, -.25}], Dashed,
Line[{sopt, .98}, {sopt, -.02}], Line[{sopt, -.09}, {sopt, -.25}],
gencol, Dotted, Line[{gmax, -0.015}, {gmax, -.25}], Line[
  {{gmin, -0.015}, {gmin, -.25}], Dashed, Line[{gopt, .98}, {gopt, -.25}]]];

ymax = 285;
dists = Plot[
  Table[PDF[NormalDistribution[ $\mu$ ,  $\sigma$ ], x] * 2000, { $\mu$ , {23, 24, 25, 26}}, { $\sigma$ , {3}}] //
  Evaluate, {x, 2, 37}, Filling → Axis, PlotRange → {{1.7, 37}, {0, ymax}},
  PlotStyle → {Lighter[Red], Red, Darker[Red], Black},
  ScalingFunctions → {Automatic, "Reverse"},
  PlotLegends → Placed[
    {" $\mu$  = 23", " $\mu$  = 24", " $\mu$  = 25", " $\mu$  = 26"}, {Scaled[{0.15, 0.5}], {0, 0.5}}],
  Frame → {False, True, True, False},
  FrameStyle → 14, FrameLabel → {None, "Frequency", None, None},
  LabelStyle → Directive[Black], ImageSize → 500, AspectRatio → 1 / 4,
  Epilog → {Text[Style["(b)", Black, 18], {3.3, -30}],
    Thick, specol, Dotted, Line[{32, 0}, {32, -ymax}],
    Line[{14.5, 0}, {14.5, -ymax}], Dashed, Line[{28, 0}, {28, -ymax}],
    gencol, Dotted, Line[{30.4, 0}, {30.4, -ymax}],
    Line[{5.8, 0}, {5.8, -ymax}], Dashed, Line[{25, 0}, {25, -ymax}]]];

TPCfig = GraphicsColumn[{TPCs, dists}, Spacings → 0]
(*Export[saveto<>"Figure2.eps", TPCfig, ImageResolution → 1000]*)

```

Out[3580]=



Appendix S2 Figure S1

This recreates the Gaussian-quadratic TPC of the organisms most similar to those used in initial models.

```
rD[T_, Topt_, sp_, CTmax_] :=  
  If[T ≤ Topt, Exp[-((T - Topt) / (2 sp))^2], 1 - ((T - Topt) / (Topt - CTmax))^2];
```

(*temporate TPCs*)

```
CTmax = 31.4; Topt = 25.3; sp = 4.8;  
CTmax2 = 28.82; Topt2 = 23.88; sp2 = 4.395;  
CTmax3 = 30.225; Topt3 = 25.8; sp3 = 4.29375;  
CTmax4 = 29; Topt4 = 24; sp4 = 4.185;  
CTmax5 = 32.1; Topt5 = 25.78; sp5 = 4.395;
```

(*tropical TPCs*)

```
TCTmax = 33.0677; TTopt = 28.8; Tsp = 2.375;  
TCTmax2 = 33.46; TTopt2 = 30.1; Tsp2 = 3.315;
```

```
TCTmax3 = 30.55; TTopt3 = 27.2; Top3 = 1.9875;
TCTmax4 = 31.78; TTopt4 = 27.98; Top4 = 3.675;
TCTmax5 = 31.96; TTopt5 = 27.84; Top5 = 3.675;
```

```
DeutschTPCs = Show[Plot[rD[T, 25, 4.8, 30.4],
  {T, 0, 37}, PlotStyle → {Darker[Cyan], Thickness[0.02]},
  PlotLegends → Placed[{"temperate TPC"}, {0.22, 0.8}],
  PlotRange → {{0, 37}, {-0.2, 1.1}}, Frame → {True, True, False, False},
  FrameStyle → 14, FrameLabel → {"Temperature, T (°C)", "Normalized Fitness r"},
  LabelStyle → Directive[Black], FrameTicks → {Automatic, {0, 0.25, 0.5, 0.75, 1}},
  ImageSize → 500, PlotRange → {-0.2, 1.1}],
Plot[rD[T, Topt, op, CTmax], {T, 0, 37},
  PlotLegends → Placed[{"Diatraea lineolata"}, {0.22, 0.75}],
  PlotStyle → Darker[Darker[Darker[Gray]]]],
Plot[rD[T, Topt2, op2, CTmax2], {T, 0, 37}, PlotLegends →
  Placed[{"Myzus persicae"}, {0.22, 0.7}], PlotStyle → Darker[Darker[Gray]]],
Plot[rD[T, Topt3, op3, CTmax3], {T, 0, 37}, PlotLegends →
  Placed[{"Aphis citricola"}, {0.22, 0.65}], PlotStyle → Darker[Gray]],
Plot[rD[T, Topt4, op4, CTmax4], {T, 0, 37},
  PlotLegends → Placed[{"Sitobion miscanthi"}, {0.22, 0.6}], PlotStyle → Gray],
Plot[rD[T, Topt5, op5, CTmax5], {T, 0, 37}, PlotLegends →
  Placed[{"H. pseudobrassicae"}, {0.22, 0.55}], PlotStyle → Lighter[Gray]],

Epilog → {{Darker[Darker[Cyan]], PointSize[0.02], Point[{30.4, 0}]},
  {Darker[Darker[Cyan]], PointSize[0.02], Point[{25, 1}]},
  {Darker[Darker[Cyan]], PointSize[0.02], Point[{5.8, 0]}},

  {Darker[Darker[Darker[Gray]]], PointSize[0.015], Point[{CTmax, 0}]},
  {Darker[Darker[Darker[Gray]]], PointSize[0.015], Point[{Topt, 1}]},
  {Darker[Darker[Darker[Gray]]], PointSize[0.015], Point[{6.1, 0]}},

  {Darker[Darker[Gray]], PointSize[0.015], Point[{CTmax2, 0}]},
  {Darker[Darker[Gray]], PointSize[0.015], Point[{Topt2, 1}]},
  {Darker[Darker[Gray]], PointSize[0.015], Point[{6.3, 0]}},

  {Darker[Gray], PointSize[0.015], Point[{CTmax3, 0}]},
  {Darker[Gray], PointSize[0.015], Point[{Topt3, 1}]},
  {Darker[Gray], PointSize[0.015], Point[{8.625, 0]}},

  {Gray, PointSize[0.015], Point[{CTmax4, 0}]},
  {Gray, PointSize[0.015], Point[{Topt4, 1}]},
  {Gray, PointSize[0.015], Point[{7.26, 0]}}]
```

```

    {Lighter[Gray], PointSize[0.015], Point[{CTmax5, 0}]},
    {Lighter[Gray], PointSize[0.015], Point[{Topt5, 1}]},
    {Lighter[Gray], PointSize[0.015], Point[{8.2, 0}]}},
    ImageSize → 350];

```

```

DeutschTPCsTrop = Show[Plot[rD[T, 28.3, 2.7, 32],
    {T, 0, 37}, PlotStyle → {Lighter[Red], Thickness[0.02]},
    PlotLegends → Placed[{"tropical TPC"}, {0.3, 0.8}],
    PlotRange → {{0, 37}, {-0.2, 1.1}}, Frame → {True, True, False, False},
    FrameStyle → 14, FrameLabel → {"Temperature, T (°C)", "Normalized Fitness r"},
    LabelStyle → Directive[Black], FrameTicks → {Automatic, {0, 0.25, 0.5, 0.75, 1}},
    ImageSize → 500, PlotRange → {-0.2, 1.1}],
    Plot[rD[T, TTopt, Top, TCTmax], {T, 0, 37},
    PlotLegends → Placed[{"Clavigralla shadabi"}, {0.3, 0.75}],
    PlotStyle → Darker[Darker[Gray]]],
    Plot[rD[T, TTopt2, Top2, TCTmax2], {T, 0, 37},
    PlotLegends → Placed[{"Stethorus punctillum"}, {0.3, 0.7}],
    PlotStyle → Darker[Darker[Gray]]],
    Plot[rD[T, TTopt3, Top3, TCTmax3], {T, 0, 37}, PlotLegends →
    Placed[{"Pterohalaeus alternatus"}, {0.3, 0.65}], PlotStyle → Darker[Gray]],
    Plot[rD[T, TTopt4, Top4, TCTmax4], {T, 0, 37},
    PlotLegends → Placed[{"Ceratitis capitata"}, {0.3, 0.6}], PlotStyle → Gray],
    Plot[rD[T, TTopt5, Top5, TCTmax5], {T, 0, 37}, PlotLegends →
    Placed[{"Diaphorina citri"}, {0.3, 0.55}], PlotStyle → Lighter[Gray]],

    Epilog → {{Darker[Darker[Red]], PointSize[0.02], Point[{32, 0}]},
    {Darker[Darker[Red]], PointSize[0.02], Point[{28.3, 1}]},
    {Darker[Darker[Red]], PointSize[0.02], Point[{17.5, 0]}},

    {Darker[Darker[Darker[Gray]]], PointSize[0.015], Point[{TCTmax, 0}]},
    {Darker[Darker[Darker[Gray]]], PointSize[0.015], Point[{TTopt, 1}]},
    {Darker[Darker[Darker[Gray]]], PointSize[0.015], Point[{19.3, 0]}},

    {Darker[Darker[Gray]], PointSize[0.015], Point[{TCTmax2, 0}]},
    {Darker[Darker[Gray]], PointSize[0.015], Point[{TTopt2, 1}]},
    {Darker[Darker[Gray]], PointSize[0.015], Point[{16.84, 0]}},

    {Darker[Gray], PointSize[0.015], Point[{TCTmax3, 0}]},
    {Darker[Gray], PointSize[0.015], Point[{TTopt3, 1}]},
    {Darker[Gray], PointSize[0.015], Point[{19.25, 0]}},

    {Gray, PointSize[0.015], Point[{TCTmax4, 0}]},
    {Gray, PointSize[0.015], Point[{TTopt4, 1]}},

```

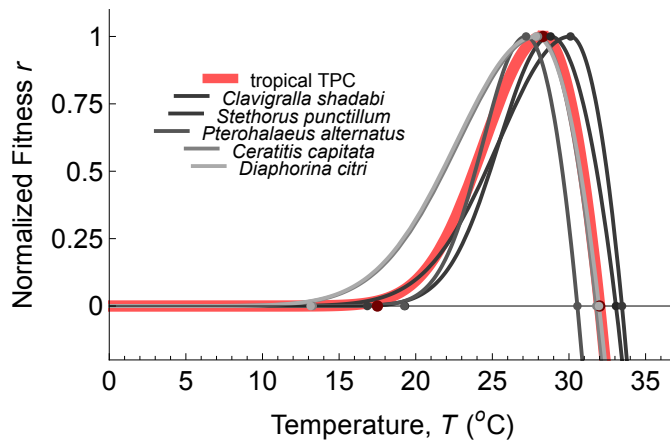
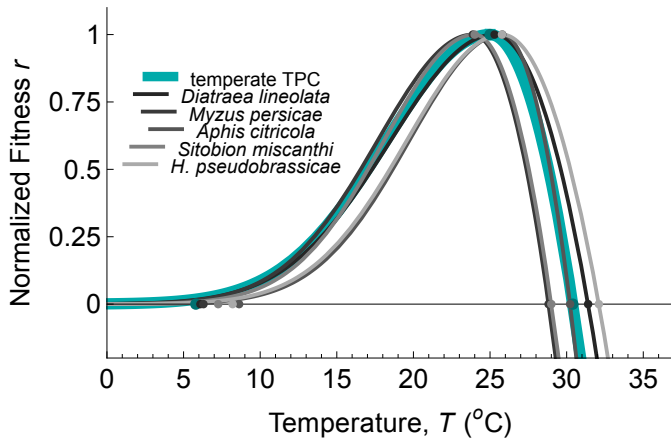
```
{Gray, PointSize[0.015], Point[{13.28, 0}]}},

{Lighter[Gray], PointSize[0.015], Point[{TCTmax5, 0}]}},
{Lighter[Gray], PointSize[0.015], Point[{TTopt5, 1}]}},
{Lighter[Gray], PointSize[0.015], Point[{13.14, 0}]}},
ImageSize → 350];
```

```
stackedTPCs = Row[{DeutschTPCs, DeutschTPCsTrop}]
```

```
Export[saveto <> "TPCsComparison_Deutschfit.pdf",
stackedTPCs, ImageResolution → 1000]
```

Out[] =



Out[] =

```
/Users/alisonrobey/Documents/PhD Year
3 (2023-2024) /Ch1/Code/TPCsComparison_Deutschfit.eps
```

Appendix S4 Figures S1 & S2

This code tests the impact of the timestep size, δt , on population dynamics. Models are stochastic, so

will not output exactly the same graphs on every run.

```
(*background functions - same as other versions with  $\delta t$  added
in and an edit to the indexing in Z to allow for multiple runs*)
r[T_, Topt_,  $\sigma$ _, CTmax_] :=
  If[T ≤ Topt, Exp[-((T - Topt) / (2  $\sigma$ )) ^ 2], 1 - ((T - Topt) / (Topt - CTmax)) ^ 2];

nt[t_, ntlast_,  $\delta t$ _, i_] :=
  
$$\frac{r[Z[[i, t]], Topt, \sigma, CTmax] * \text{Exp}[r[Z[[i, t]], Topt, \sigma, CTmax] * \delta t]}{\left(\frac{r[Z[[i, t]], Topt, \sigma, CTmax]}{ntlast} - \alpha\right) + \alpha \text{Exp}[r[Z[[i, t]], Topt, \sigma, CTmax] * \delta t]}$$
;

(*unchanging parameters for TPC and temps*)
Topt = 25;  $\sigma$  = 5; CTmax = 30; (*TPC params*)
 $\alpha$  = .001;
 $\mu T$  = 25;  $\sigma T$  = 3; (*temp params-values for some extinction risk*)

tmax = 100; (*number of time steps*)
nmax = 500; (*number of replicates*)

(*initializing*)
 $\delta ts$  = Join[Range[0.01, 0.09, 0.01],
  Range[0.1, 0.9, 0.1], Range[1, 10], Range[20, 100, 10]];

means = vars = stdevs = ConstantArray[0, {nmax, Length[ $\delta ts$ ]}];
cvs = equidists = ConstantArray[0, Length[ $\delta ts$ ]];
extTime = ConstantArray[tmax, {nmax, Length[ $\delta ts$ ]}];

(*set up thermal regimes*)
Z = ConstantArray[0, {nmax, tmax}];
Do[Z[[i, k]] = RandomVariate[NormalDistribution[ $\mu T$ ,  $\sigma T$ ]],
  {k, 1, tmax}, {i, 1, nmax}];

 $\mu r$  = ConstantArray[0, nmax];
Do[ $\mu r$ [[i]] = Mean[Table[r[Z[[i, k]], Topt,  $\sigma$ , CTmax], {k, 1, tmax}]], {i, 1, nmax}];

ninit = ConstantArray[0, nmax];
Do[ninit[[i]] =  $\mu r$ [[i]] /  $\alpha$ , {i, 1, nmax}];

Ks = ConstantArray[0, {nmax, tmax}];
Do[Ks[[i, k]] = r[Z[[i, k]], Topt,  $\sigma$ , CTmax] /  $\alpha$ , {k, 1, tmax}, {i, 1, nmax}];

p = ConstantArray[0, {nmax, Length[ $\delta ts$ ], tmax}];
Do[p[[i, All, 1]] = ninit[[i]], {i, 1, nmax}];
```

```

(*simulations*)
Do[p[[i, j, k]] = nt[k, p[[i, j, k - 1]],  $\delta$ ts[[j]], i];
  If[p[[i, j, k]] < ((1/ $\alpha$ ) * 10^-6), extTime[[i, j]] = k(*;Break[;*)],
  {k, 2, tmax}, {j, 1, Length[ $\delta$ ts]}, {i, 1, nmax}]

(*calculate stats*)
Do[If[extTime[[i, j]] == tmax,

  means[[i, j]] = Mean[p[[i, j]]];
  vars[[i, j]] = Variance[p[[i, j]]];
  stdevs[[i, j]] = StandardDeviation[p[[i, j]]],

  means[[i, j]] = Mean[Take[p[[i, j]], extTime[[i, j]]];
  vars[[i, j]] = Variance[Take[p[[i, j]], extTime[[i, j]]];
  stdevs[[i, j]] = StandardDeviation[Take[p[[i, j]], extTime[[i, j]]];

], {j, 1, Length[ $\delta$ ts]}, {i, 1, nmax}]

allmeans = Mean[means]; allvars = Mean[vars]; allstdevs = Mean[stdevs];
meanplot = Transpose[{ $\delta$ ts, allmeans}];
varplot = Transpose[{ $\delta$ ts, allvars}];
stdevplot = Transpose[{ $\delta$ ts, allstdevs}];

ListLogLinearPlot[meanplot, PlotLabel → "Means"];
ListLogLinearPlot[varplot, PlotLabel → "Vars"];
ListLogLinearPlot[stdevplot, PlotLabel → "Stdevs"];

Do[cvs[[j]] = allstdevs[[j]] / allmeans[[j]],
  {j, 1, Length[ $\delta$ ts]}];
cvplot = Transpose[{ $\delta$ ts, cvs}];
ListLogLinearPlot[cvplot, PlotLabel → "CVs"];

dist = ConstantArray[0, {nmax, Length[ $\delta$ ts], tmax}];
Do[dist[[i, j, k]] = Abs[Ks[[i, k]] - p[[i, j, k]],
  {k, 1, tmax}, {j, 1, Length[ $\delta$ ts]}, {i, 1, nmax}];
meandist = ConstantArray[0, {nmax, Length[ $\delta$ ts]}];
Do[meandist[[i, j]] = Mean[dist[[i, j]]], {j, 1, Length[ $\delta$ ts]}, {i, 1, nmax}];
meandists = Mean[meandist];
meandistplot = Transpose[{ $\delta$ ts, meandists}];
ListLogLinearPlot[meandistplot, PlotLabel → "MeanDists"];

timestepplot = Column[{
  Show[ListLogLinearPlot[varplot, Joined → True,

```

```

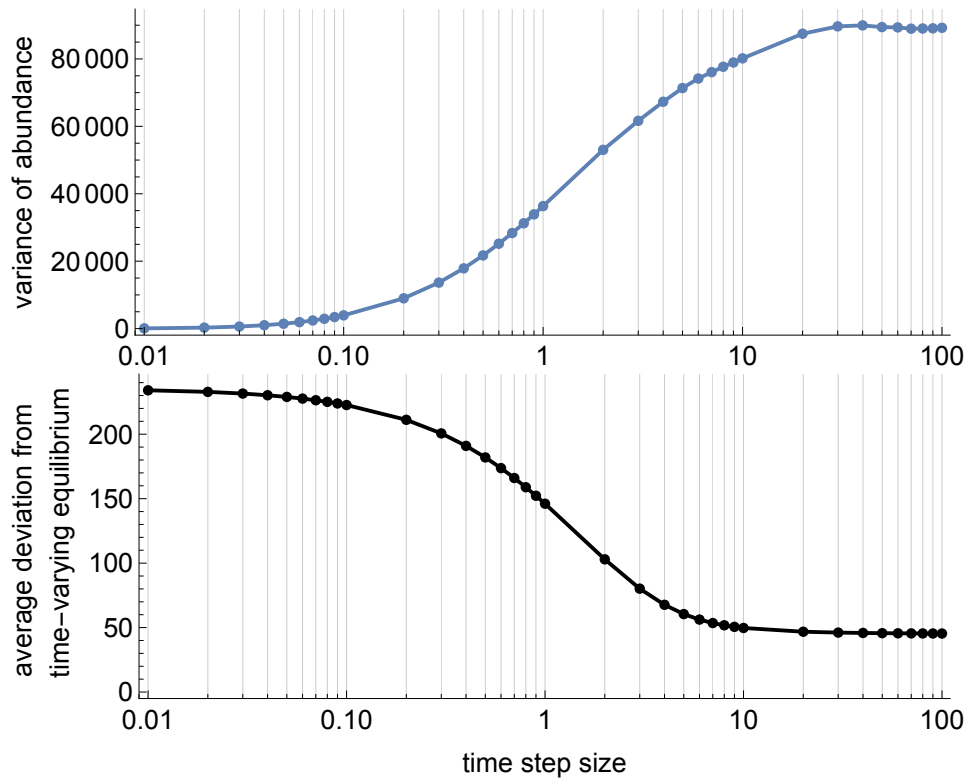
    PlotRange → {{0.009, 110}, Automatic}, GridLines → { $\delta$ ts, None},
    Frame → {True, True, False, False},
    FrameStyle → 14, FrameLabel → {None, "variance of abundance"},
    AspectRatio → .4, ImageSize → 500], ListLogLinearPlot[varplot]],
Show[ListLogLinearPlot[meandistplot, PlotStyle → Black, Joined → True,
    PlotRange → {{0.009, 110}, Automatic}, GridLines → { $\delta$ ts, None},
    Frame → {True, True, False, False}, FrameStyle → 14, FrameLabel →
        {"time step size", "average deviation from \n time-varying equilibrium"},
    AspectRatio → .4, ImageSize → 500], ListLogLinearPlot[
    meandistplot, PlotStyle → Black]], Spacings → {0, 0}]

colorFunc = ColorData["SolarColors"];
run = 15;
total = 9;
line = ConstantArray[ninit[[run]], tmax];
timestepseriesplot = Show[
    ListLinePlot[line, PlotRange → {{10, 30}, {0, 1000}}, PlotStyle → {Black, Dotted},
        PlotLegends → {"K"}, Frame → {True, True, False, False},
        FrameStyle → 14, FrameLabel → {"time step", "Population Size"}],
    ListLinePlot[p[[run, 1]],
        PlotStyle → colorFunc[9 / total], PlotLegends → {" $\delta$ t = 0.01"}],
    ListLinePlot[p[[run, 5]],
        PlotStyle → colorFunc[8 / total] (*, PlotLegends → {" $\delta$ t = 0.05"} *)),
    ListLinePlot[p[[run, 10]],
        PlotStyle → colorFunc[7 / total], PlotLegends → {" $\delta$ t = 0.1"}],
    ListLinePlot[p[[run, 14]],
        PlotStyle → colorFunc[6 / total] (*, PlotLegends → {" $\delta$ t = 0.5"} *)),
    ListLinePlot[p[[run, 19]],
        PlotStyle → colorFunc[5 / total], PlotLegends → {" $\delta$ t = 1"}],
    ListLinePlot[p[[run, 23]],
        PlotStyle → colorFunc[4 / total] (*, PlotLegends → {" $\delta$ t = 5"} *)),
    ListLinePlot[p[[run, 28]],
        PlotStyle → colorFunc[3 / total], PlotLegends → {" $\delta$ t = 10"}],
    ListLinePlot[p[[run, 32]],
        PlotStyle → colorFunc[2 / total] (*, PlotLegends → {" $\delta$ t = 50"} *)),
    ListLinePlot[p[[run, 37]],
        PlotStyle → colorFunc[1 / total], PlotLegends → {" $\delta$ t = 100"}],
    ListLinePlot[Ks[[run]],
        PlotStyle → {Black, Dashed}, PlotLegends → {Subscript["K", "i"]}],
    ImageSize → 500]

Export[saveto <> "timestepplot.pdf", timestepseriesplot, ImageResolution → 1000]
Export[saveto <> "timestepseriesplot.pdf", timestepplot, ImageResolution → 1000]

```

Out[*]=



Out[*]=

